

BMJ Open A systematic review of the scope and impact of rural primary healthcare innovations using digital health technology

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ABSTRACT

Objectives Digital technology in primary healthcare service delivery can enhance accessibility, service delivery and health outcomes in rural populations. The objective of this systematic review is to review and synthesise the scope and impact of digital health technology innovations within rural primary healthcare settings.

Design Systematic review.

Data sources Articles published on PubMed, PsycINFO, Cochrane Central, SCOPUS, Web of Science, EMBASE and CINAHL between January 2013 and October 2025 were searched using key search terms.

Eligibility criteria Patient, intervention, context, outcome model criteria guided article eligibility. Included articles were undertaken in rural populations, used digital health technology for treatment or management, explored the impact of digital health technology on rural primary healthcare and reported on healthcare outcomes. Included articles were in the English language and presented peer-reviewed primary research.

Data extraction and synthesis Extraction was performed using a bespoke standardised template by multiple reviewers. Quality assessment was undertaken using the Mixed Methods Appraisal Tool. Descriptive analysis and conventional inductive content analysis were applied to quantitative and qualitative data, respectively. The review is written in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols statement guidelines.

Results 66 studies were included in the review. Most studies were conducted in the USA (n=26). Most studies focused on adult patient populations, with limited representation of Indigenous (n=3) and paediatric populations (n=2). Telemedicine/telehealth interventions using audio, video or both were the most common (n=36). Remote patient monitoring or point-of-care testing was integrated into 21 studies. Physical health conditions, particularly diabetes (n=17), cardiovascular diseases (n=11) and general primary healthcare concerns (n=13) were commonly reported. Others reported on areas including mental health, hypertension, obesity and pregnancy care.

Conventional inductive content analysis identified key themes: cost and time effectiveness, quality healthcare provision, consumer acceptance from both patients

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ This study provides a comprehensive review of digital health technology used in rural primary healthcare from studies across the globe and diverse health fields.
- ⇒ The broad definitions used for digital health technology captured an expansive variety of approaches to improving rural primary healthcare.
- ⇒ Descriptive and content analysis methods have been used for analysis, as the diversity of included articles precludes the application of meta-analysis and thematic analysis.
- ⇒ The search strategy inadvertently failed to capture more than a non-trivial sample of artificial intelligence-based interventions.
- ⇒ This study is strengthened by its inclusion of diverse geographies, methodological approaches and digital health technologies; however, care must be taken in generalising its findings to new contexts due to inherent differences in international healthcare systems.

and practitioners, and healthcare service provider perspectives. Uptake barriers included staff workload and patient non-compliance, while facilitators encompassed process standardisation and practitioner acceptance and endorsement. Consumer acceptance was linked to satisfaction, willingness to engage and improved health outcomes and well-being.

Conclusions Digital health interventions in rural primary healthcare offer significant potential to improve healthcare delivery, reduce costs and enhance patient access, satisfaction and health outcomes. However, careful consideration of factors such as feasibility, consumer and practitioner acceptance, and recognition of limitations is crucial for successful implementation. The review underscores the importance of flexible policies to support emerging digital healthcare solutions, including the integration of artificial intelligence. Overall, digital health interventions offer a promising avenue to improve healthcare outcomes in rural areas and should be prioritised for government funding and investment.

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INTRODUCTION

Globally, many rural and remote populations, even in advanced economies, experience significant inequalities in primary healthcare (PHC) access and outcomes. For example, rural residents in Australia and the USA face fewer healthcare services and higher mortality rates, increased smoking prevalence, lower household income, ageing population and a greater burden of disease than their urban counterparts.^{1–3} Barriers to healthcare access disproportionately impact rural residents. Rural residents have greater reluctance to access care due to cultural and financial obstacles,⁴ greater travel distance and limited transport options,⁵ reduced availability of services and a lack of suitably trained local staff.⁶ Delivering healthcare services in rural areas is also complex from a provider perspective. Some services can become economically strained as rural patients—often with lower incomes—may forgo consultations, diagnostic tests and medications.⁷ Recruiting and retaining qualified healthcare providers in rural areas is also challenging, with many providers reporting burnout due to high workloads and insufficient staffing.^{6,7}

A ‘best fit’ framework synthesis⁸ using the PHC access framework also identified several larger structural barriers to delivering services in rural areas. Such barriers included lower salaries and higher turnover for healthcare staff, lack of access to health infrastructure (eg, pharmacies and mobile health clinics) and inadequate supply of medicines and medical equipment, reduced operating hours, facilities which are not disability accessible and long patient wait times.⁹ In some low-/middle-income countries, internet access can be limited by geography and cost. Even in wealthy countries, rural areas may face limited connectivity due to factors such as geographic isolation (Australia), unequal infrastructure distribution (Canada) and lower household income (the USA).¹⁰

In 2020, the WHO developed a digital health strategy with the aim of enhancing the development and implementation of sustainable digital health solutions.¹¹ This global strategy recognises the need to address health inequities and healthcare worker shortages in rural locations. Digital health technology enables monitoring of patient health status, timely interventions and informed management plans.¹² Digital health technology is highly diverse and encompasses a wide range of applications, from wearable sensor systems to telephone consultations, point-of-care testing devices and virtual health consultations with specialists.¹³

Digital health technology has the potential to address some of the barriers to rural populations accessing healthcare and improving health outcomes. However, a thorough assessment is required to ensure these technologies can achieve improvements in healthcare amidst substantial barriers. Despite numerous digital health interventions being trialled in rural communities worldwide, an overarching analysis of their role within rural PHC is lacking.

Aims

This systematic review aimed to review and synthesise global empirical evidence on the scope and impact of digital health technology innovations within PHC services for rural populations.

METHODS

This systematic review protocol was registered with the PROSPERO database and has been previously published.¹⁴ The reporting in this review is aligned with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses protocol for systematic reviews.¹⁵

Given the diverse geographical landscapes worldwide, a standardised global classification system for ‘rural’ is not feasible. For simplicity, included studies which were defined by their authors as occurring in remote, rural or regional areas are termed ‘rural’ in this review. Similarly, studies reporting digital health innovation or technology were included based on their individual author definitions and according to the review selection criteria. The scope of this literature review was also deliberately confined to examining the delivery and implementation of virtual care, excluding studies focused primarily on data use, analytics or secondary digital functions.

Search strategy

A systematic search was conducted across title and abstract fields using seven electronic bibliographic databases: PubMed, PsycINFO, Cochrane Central, SCOPUS, Web of Science, EMBASE and CINAHL. Search strategies were adjusted according to the requirements of each database. Initially, a pilot search was conducted using a combination of both general and specific search terms to identify different key terms for possible inclusion, before finalising the selection. A test search of each term was performed by removing the keyword, re-running the search and comparing the results. Keywords that did not alter the number of records returned were considered redundant and excluded from the final search string. The final search strategy was then applied across all databases.

Results were restricted to peer-reviewed articles in English, published between January 2013 and December 2023. The initial search was conducted in February 2024 and finalised in March 2024. A full search strategy and MeSH terms have previously been published.¹⁴ The full search strategy for each bibliographic database is also provided in online supplemental material 1.

Eligibility criteria

To be eligible for inclusion, studies had to meet the PICO (Patient, Intervention, Context, Outcome) model criteria with specific exclusion and inclusion parameters, as described in [table 1](#). Primary research of any study design (ie, cross-sectional, longitudinal, survey, experimental, programme evaluation, qualitative or mixed methods) that included outcomes analysis was eligible for inclusion. Research designs included qualitative, quantitative

Table 1 Eligibility criteria

PICO	Inclusion criteria	Exclusion criteria
Population	Any study conducted with rural populations, including studies where rural populations form only a part of the full study	Any studies without any rural populations
Investigated phenomena	Studies conducted within the primary healthcare sector and describing digital health technology aimed at disease treatment or management. This includes studies such as telehealth, remote patient monitoring, virtual healthcare, wearable devices and point of care testing	Studies not focused within the primary healthcare sector or not using digital health technologies aimed at patient treatment or management. This includes studies describing any health information sharing systems/platforms, for example, patient record systems or electronic health records
Context	Studies demonstrating the scope and impact of digital health technology within rural primary healthcare in improving patient healthcare access and outcomes	Studies outside the included context, such as hospital settings and metropolitan healthcare services
Outcome	Improved rural primary healthcare outcomes, including clinical outcomes, technology acceptability, and barriers and enablers to care	No rural primary healthcare outcomes
Study design	Primary research studies, including both quantitative studies (randomised controlled trials, cohort studies, case-control studies, survey studies) and qualitative studies (mixed-methods studies, focus groups, interviews)	Secondary research, including systematic reviews, editorials, opinion pieces, commentaries, position papers, conference abstracts or posters, protocols or theses
Time frame	1 January 2013 until 29 October 2025	Studies prior to 2013
Other	English language studies Peer-reviewed studies	Non-English studies Not peer-reviewed studies

or mixed-methods studies. Qualitative studies included interviews, open-ended surveys, participant observations or focus groups. Mixed-method studies were considered if data from the quantitative or qualitative components were extractable. Secondary research, such as systematic reviews or meta-analyses, was excluded. Similarly, commentaries, perspectives, letters, reviews, editorials or opinion pieces, or grey literature were excluded.

Study selection

Citations were transferred to EndNote V.20 and uploaded into Covidence for deduplication, screening and data management. Abstracts and titles were screened independently by a minimum of two reviewers, ensuring selected studies met the inclusion criteria. Any discrepancies were discussed, and conflicts were reviewed by a third reviewer to reach a consensus. Of the remaining studies, full-text articles were screened independently by two reviewers (PG, CW, WM). Conflicts at this stage were cross-checked and discussed between the reviewers to reach a consensus.

Data extraction

Four reviewers deductively designed a data extraction template with the assistance of a research librarian to standardise the data extraction process (CW, PG, BFN, WM). This template extracted data, including country, aim, study design, participant composition, digital intervention type, intervention duration, comparator and data relating to PHC outcomes. These outcomes were kept

broad (in order to enable a more inductive analysis) and included clinical outcomes, acceptability, barriers and enablers to care, and care quality.

Three reviewers independently extracted a subset of articles from the final list of screened articles and then further refined the data extraction template prior to completing full extraction (CW, PG, WM). A fourth reviewer examined the updated data extraction template and the extracted data for accuracy and completeness (BFN). Studies deemed ineligible or with incomplete information during data extraction were excluded.

Quality assessment

Three reviewers independently assessed each paper's quality using the Mixed Methods Appraisal Tool (MMAT) (CW, PG, WM).^{16 17} Designed with systematic appraisal efforts in mind, this tool facilitates assessments of systematic mixed studies, including qualitative, quantitative and mixed-methods research. Scoring was based on 12 criteria distributed across the following domains: (1) description of aims and objectives, (2) description of methods, (3) participant selection, (4) data collection, (5) data analysis, (6) reporting and (7) engagement. Based on these criteria, studies were identified as being of good or poor quality. The MMAT has been shown to have high validity¹⁶ and reliability,¹⁸ making it a useful tool for appraising literature reviews.¹⁷ The final scoring of all studies was cross-checked and a consensus was achieved by discussion.

Analysis and synthesis

Given the broad inclusion criteria of this systematic review, both descriptive analysis and conventional inductive content analysis were used for data analysis. Descriptive analysis was applied to extracted generalisable quantitative data, such as studies reporting changes in HbA1c. Extracted qualitative data (including programme evaluation data reported in included studies which were otherwise quantitative) were analysed using inductive content analysis in accordance with previously described methods.^{19 20} The initial processing of reading and generating tentative codes was undertaken by one reviewer (WM) and further refined by two reviewers (CW, PG). Final codes and categories were agreed on by all authors.

RESULTS

Quality assessment

Quality assessment using MMAT indicated that almost all studies were of good methodological quality (n=66) (online supplemental material 2).

Results summary

Initial searches identified 6295 studies (figure 1). After removing 1820 duplicates, 4475 studies remained for title and abstract screening, which resulted in 206 studies deemed eligible for full-text review. Screening resulted in 66 eligible studies included in this review. A basic description of these studies is provided in table 2, due to formatting requirements, full study details are

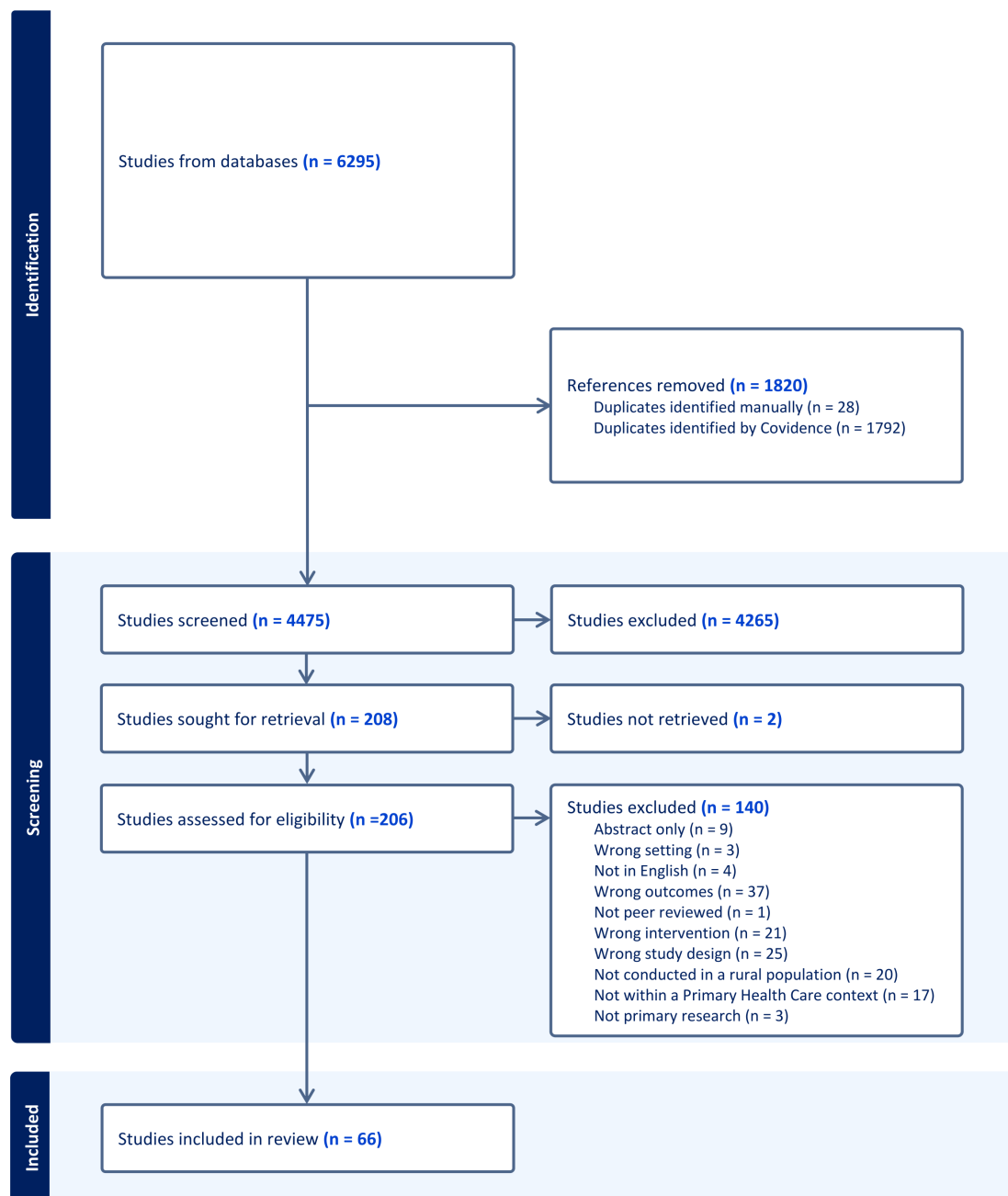


Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses chart of study search and selection process.

Table 2 Basic study characteristics

Citation	Country	Study design
Beks <i>et al</i> ⁴⁹	Australia	Mixed methods
Doogue <i>et al</i> ³⁸	Ireland	Mixed methods
Kirkland <i>et al</i> ²¹	USA	Quantitative non-randomised (before and after)
Rodríguez Sánchez-Laulhé <i>et al</i> ⁵⁸	Spain	Pragmatic randomised control trial
Siminerio <i>et al</i> ²²	USA	Non-randomised trial
Verma <i>et al</i> ⁴⁸	India	Crossover randomised control trial
Zullig <i>et al</i> ⁶⁹	USA	Mixed methods
Feldmeier <i>et al</i> ⁵⁹	Germany	Explorative qualitative
Haggerty <i>et al</i> ⁵¹	USA	Analytical cross-sectional study
Kobe <i>et al</i> ²³	USA	Mixed methods
Meng <i>et al</i> ⁶⁵	China	Non-comparative study (quantitative descriptive)
Camargo <i>et al</i> ²⁴	Brazil	Non-comparative study (quantitative descriptive)
de Batlle <i>et al</i> ⁵⁵	Spain	Prospective pragmatic, two-arm parallel implementation trial – integrated care arm vs usual care
Gong <i>et al</i> ⁶⁰	China	Clustered randomised control trial
Kwak <i>et al</i> ²⁵	Korea	Quasi-experimental
Yan <i>et al</i> ⁶⁴	China	Clustered randomised control trial
Lu <i>et al</i> ²⁶	USA	Quasi-experimental
Brown <i>et al</i> ⁷⁰	USA	Analytical cross-sectional study
Martin <i>et al</i> ⁷⁸	Spain	Qualitative descriptive
Orchard <i>et al</i> ⁶²	USA	Analytical cross-sectional study
Patel <i>et al</i> ³⁷	Indonesia	Quasi-experimental
Peiris <i>et al</i> ⁴¹	India	Clustered randomised control trial (stepped wedge)
Shah <i>et al</i> ⁶⁶	India	Mixed methods
McLendon <i>et al</i> ²⁷	USA	Pre and post
Vedanthan <i>et al</i> ⁷⁹	Kenya	Clustered randomised control trial
Bonnell <i>et al</i> ⁷¹	Dominican Republic	Non-comparative study (quantitative descriptive)
Chang <i>et al</i> ⁸⁰	USA	Non-comparative study (quantitative descriptive)
Chauhan <i>et al</i> ³⁶	India	Non-randomised control trial
Gurbeta <i>et al</i> ⁸¹	Bosnia and Herzegovina	Non-comparative study (quantitative descriptive)
Mallow <i>et al</i> ⁵⁴	USA	Pre and post
Scogin <i>et al</i> ⁶⁷	USA	Randomised control trial
Barrera <i>et al</i> ⁷⁴	USA	Case studies
Shetty <i>et al</i> ⁸²	India	Non-comparative study (quantitative descriptive)
Evangelista <i>et al</i> ⁸³	Spain	Descriptive cross-sectional study
Smith and Satyshur ²⁸	USA	Qualitative descriptive
Davidson <i>et al</i> ⁶³	Sweden	Randomised control trial
Gonzalez and Brossart ⁸⁴	USA	Time series+case studies
Richter <i>et al</i> ⁶¹	USA	Randomised control trial
Tian <i>et al</i> ⁴⁰	China and India	Clustered randomised control trial
Shane-McWhorter <i>et al</i> ²⁹	USA	Time series with comparison group
Siminerio <i>et al</i> ³⁰	USA	Pre and post
Davis <i>et al</i> ⁷³	USA	Randomised control trial
Fortney <i>et al</i> ⁷⁵	USA	Randomised pragmatic comparative effectiveness trial

Continued

Table 2 Continued

Citation	Country	Study design
Reid <i>et al</i> ⁵⁷	Australia	Clustered randomised control trial
Nasir <i>et al</i> ⁵³	Australia	Decolonising methodologies for co-design consensus and a qualitative interpretive-descriptive thematic analysis framework
Maulik <i>et al</i> ⁸⁵	India	Parallel group cluster randomised usual care-controlled trial
Wańczura <i>et al</i> ⁵⁶	Poland	Prospective, non-controlled study
Alcoceba-Herrero <i>et al</i> ⁸⁶	Spain	Prospective, multicentre, open-label randomised clinical trial
Calderon <i>et al</i> ⁵²	Philippines	Qualitative cross-sectional study
Charly and Swedlund ⁵⁰	USA	Qualitative descriptive
Chen <i>et al</i> ⁸⁷	China	Observational study post cluster RCT
Eberly <i>et al</i> ⁸⁸	USA	Stepped-wedge, pragmatic comparative effectiveness trial
Egede <i>et al</i> ³¹	USA	Randomised controlled trial
Hickey <i>et al</i> ⁸⁹	Kenya, Uganda	Randomised controlled trial
Leonard <i>et al</i> ³²	USA	Mixed methods
Mackwood <i>et al</i> ³³	USA	Retrospective cohort study
Maharani <i>et al</i> ⁴⁵	Indonesia	Retrospective cohort study
Marshall <i>et al</i> ⁶⁸	USA	Qualitative descriptive
Siedner <i>et al</i> ⁹⁰	South Africa	Open-label, randomised, controlled trial
Zhang <i>et al</i> ³⁴	China	Open-label, parallel, cluster randomised controlled trial
Lamas <i>et al</i> ⁴⁷	Brazil	Multicentre prospective design
Herbert <i>et al</i> ⁴³	Australia	Parallel group cluster randomised control trial
Norberg <i>et al</i> ⁴⁶	Norway	Qualitative descriptive
Schumacher <i>et al</i> ⁴²	Australia	Randomised controlled trial
Schumacher <i>et al</i> ⁴⁴	Australia	Randomised controlled trial
Zhang <i>et al</i> ³⁹	China	Secondary analysis of a cluster randomised controlled trial

presented in online supplemental material 3. Of these, 52 were quantitative, 9 were qualitative and 5 were mixed methods studies. Among the quantitative studies, randomised control trial designs were the most common (n=27), followed by before-and-after designs (n=7), by non-comparative studies using descriptive statistics (n=6), quasi-experimental designs (n=6), retrospective cohort studies (n=4) and analytical cross-sectional studies (n=2). Sample sizes ranged broadly, largely due to methodology, from n=3 in a qualitative study to n=46 520 in a study using retrospective data. Descriptive analysis of quantitative data and the results of a conventional inductive content analysis are now presented.

Study characteristics

66 studies were included in this review. Details of these studies and their findings are presented in online supplemental material 4. Studies were predominantly based in the USA (n=26). Multiple studies were undertaken in India (n=7), Australia (n=6), China (n=7), Spain (n=5) and Kenya (n=2). Other studies were identified from countries located in Europe, Africa, Asia, South America and the Caribbean.

Participants were predominantly adult patients receiving healthcare (henceforth 'patients'), except

for one study which recruited caregivers of paediatric patients with diabetes. Several studies (n=11) also included input from health professionals. Most studies focused on general adult patient populations, with Indigenous (n=3) and paediatric populations (n=2) less well represented. Some studies (n=5) focused on American military veterans.

The majority of studies reported the use of telemedicine or telehealth intervention technologies using either audio, video or both types of technology (n=36). Studies also reported the use of mobile or web-based applications as part of the intervention (n=20). Remote patient monitoring technology or point-of-care testing was also integrated into many studies (n=21).

Studies mostly addressed health conditions such as diabetes (n=17), cardiovascular diseases (CVDs) (n=11) and general PHC concerns (n=13). A descriptive analysis of findings related to these most investigated areas is provided in the subsequent sections. Other areas investigated included depression and dementia (n=7), hypertension (n=4), obesity (n=2), stroke (n=2), heart failure (n=2), pregnancy care (n=1), infant care (n=1) and ear disease (n=1). For the sake of brevity, these clinical findings are not further explored in this review. A summary of

all reported findings is presented in online supplemental material 4.

Diabetes mellitus

Studies commonly reported the use of digital technology to manage diabetes (ie, ICD-10 classifications (E08-E13)).^{21–35} These studies highlighted the effectiveness of digital health technologies in diagnosing, treating and managing diabetes in rural populations. A number of studies found statistically significant reductions in HbA1c in their interventions.^{21–23 27 29} Other reported benefits to patients included statistically significant improvements to: anxiety, depression and medication adherence,²⁵ diabetes-related emotional stress,³⁰ achievement of low-density lipoprotein goals,²⁹ diabetes self-management education and hospitalisation frequency²⁷ and knowledge of diabetes.²⁹

Cardiovascular disease

Several studies explored the impact of their interventions on patients with CVD. One study showed improved times to, and rates of, aspirin administration in the intervention compared with control.³⁶ Others reported statistically significant reductions in systolic blood pressure which favoured the intervention groups³⁷ and likely reductions in systolic and diastolic blood pressure in the intervention group (statistical analysis was not feasible in the study).³⁸ A study of atherosclerotic CVD reported significant reductions in systolic and diastolic blood pressures in the intervention groups compared with usual care.³⁹ Similarly, one study undertaken in both India and China reported reductions in systolic blood pressure, but only in Chinese intervention groups.⁴⁰ This outcome was repeated in another Indian-based study which further demonstrated no impact from their intervention on blood pressure control.⁴¹ Others reported no improvement in total cholesterol, LDL cholesterol or blood pressure⁴² or non-high density lipoprotein cholesterol or obesity prevalence in intervention vs control groups.³⁹ Three publications (all from the same core study) identified significant improvements in outcomes, including total energy intake from nutrient-dense core foods, added sodium consumption, quality of life, patient activation, HbA1c and body weight which favoured their intervention groups.^{42–44} Two studies showed long-term improvements, including a significantly reduced 10-year and lifetime risk of atherosclerotic CVD in the intervention group and all-cause mortality in their intervention groups.^{39 45}

General Primary Healthcare

Telehealth was explored by many studies as an avenue for managing general PHC concerns. Studies identified that telehealth was broadly suitable for primary care, particularly where patients presented with concerns that were simple and presented few issues or risks.⁴⁶ However, in more complex situations, telehealth was seen by general practitioners as presenting a risk of missing important information which would likely be detected during

in-person consultations.⁴⁶ In contrast to these reservations, two other studies identified that telehealth appointments had high problem resolution rates (85%)⁴⁷ and high diagnostic (74%) and treatment (80%) concordance when compared with in-person care: particularly for the treatment of hypertension (95%, Cohen kappa=0.93) and diabetes (93%, Cohen kappa=0.89).⁴⁸ Telehealth approaches were reported by patients to be well suited to appointments focused on medication checks, mental health problems and disease follow-ups, but less suited to addressing issues relating to physical health, wellness and new problems.^{49 50} Telehealth also supports patients to share health concerns, receive reassurance and feel connected to care,⁴⁹ and was seen as a valuable tool for developing relationships with patients who would otherwise avoid in-person care.⁴⁶ For instance, patients with anxiety, depression and endocrinological conditions requiring monitoring were found to be more likely to use telehealth.⁵¹ Significant improvements in anxiety and depression (2.25, 95% CI 1.14 to 4.45; $p<0.019$) and medication adherence (OR=1.76, 95% CI 1.34 to 2.31; $p<0.001$) were also reported in a telehealth intervention which connected patients to specialists via telehealth appointments facilitated in person by local PHC nurses.²⁵

Several studies used applications (apps) in general practice to improve patient care. One study deployed a practitioner-facing app to support practitioners with medical calculations, drug interactions, patient education and latest evidence-based medical information. This tool empowered practitioners to deliver evidence-based care, particularly non-medical staff such as midwives and community health workers. Using the app, these staff were able to diagnose and treat patients (with later medical review) in circumstances where medical staff were unavailable.⁵² A number of studies used apps which focused on self-monitoring at home. By enabling regular collection of patient data, these applications provided the ability for general practitioners to better understand their patients (eg, their ongoing functioning, mental health, physical health) and assist them to reach a diagnosis or manage their conditions.⁵³ App-based interventions supporting self-management reported significantly improved blood glucose, systolic blood pressure and diastolic blood pressure ($p<0.001$), body mass index ($p=0.004$)⁵⁴; significantly improved quality of life indicators,^{55 56} improved subjective health status,⁵⁶ reductions in unplanned healthcare visits ($p=0.004$),⁵⁵ improved emotional self-awareness ($\beta=0.59$, $p=0.048$)⁵⁷ and improved pain, hand function and upper limb function in patients with osteoarthritis.⁵⁸

Conventional content analysis

The thematic synthesis describes how digital health technology has been used and integrated into the primary care services of disparate rural populations across the globe. The analysis identified three main categories: (1) cost and time effectiveness, (2) quality healthcare provision, and (3) consumer acceptance.

Cost and time effectiveness

For patients

Several studies reported the broader benefits that digital health interventions brought to patients. Interventions substantially reduced travel costs and time required to receive healthcare.^{10 27 28 49 59} Flow-on impacts from reduced travel time included reductions to time taken off work for patients and/or their carers while accessing healthcare.⁵⁹ Three studies noted that additional financial factors, such as phone credit and internet expenses, internet service and challenges using technology, limited the effectiveness of the intervention for some patients.^{32 49 60} One study—in which the intervention group travelled to their primary care clinic for integrated telemedicine and the control group participated in phone counselling—reported that the intervention increased time burdens and travel costs for patients.⁶¹

For practitioners/health service

Health practitioners and services were found to generate savings from using digital health interventions. These savings were due to avoided strokes,⁶² and as general savings on a dollar-per-dollar or per-patient basis.^{27 55 63} Several studies described digital health intervention's operating costs as a 'low cost' per patient.⁶⁴ One study described savings on staff travel due to their digital health intervention usage.⁶³ As in the previous subtheme, one study reported substantially increased provider costs for each patient who successfully quit smoking compared with usual care.⁶¹

Quality healthcare provision

Practitioner perspective

Few studies directly questioned practitioners about their views on the interventions. Studies reported that most professionals thought the interventions were important,³⁵ helpful,⁶⁵ improved appointment attendance,⁵¹ supported their work,⁶⁶ convenient, increased confidence and empowered clinical decision making,⁵² were effective for engaging hesitant and vulnerable patients,⁴⁶ and increased effectiveness.⁴⁸ Other findings included that health professionals would continue to refer to the intervention,²⁷ that relationships between practitioners, patients and family members improved, and that interventions were suitable for follow-up care,⁵⁹ and that therapeutic alliance scores were similar to other interventions.⁶⁷ Participation in interventions improved the frequency of contact between patients and health professionals, improving awareness and accountability regarding their conditions.²³ Improved frequency of contact may contribute to significant reductions in unplanned healthcare system use and emergency department visits and inpatient hospitalisations,^{27 55} and lessened need for in-person counselling from accredited social health activists.⁶⁶

Limitations were discussed in a few studies. Some concerns were raised in studies using telehealth interventions. These studies suggested practical limitations

in information-gathering capabilities, which could sometimes result in failure to identify incidental findings.^{46 48 49} Another study noted negative feedback regarding the quality of information provided to patients in their intervention.³⁵

Access

Digital health interventions improved access to care for patients in rural and remote communities. Improved access was attributed to point-of-care devices,^{24 53} clinical decision support tools⁵² and telehealth services.^{28 47 49 51 68} Patients also benefited from improved access to care due to increased care coordination and healthcare system navigation,⁶⁹ with some interventions potentially reducing educational and economic barriers to accessing care.⁵¹ Reductions in overall need for healthcare utilisation by patients may lead to corresponding improvements in service accessibility.^{27 55 66}

Feasibility

The most common barrier to feasibility mentioned was staff workload. Studies reported mixed impacts on workload, with some describing increased workload,²³ unchanged workload^{22 68} or increased workload for some staff (ie, general practitioners) but not others (ie, specialists).⁵⁹ Other barriers included staff shortages⁷⁰ and pre-existing heavy workloads on the effectiveness of interventions.⁶⁰ Patient non-compliance with interventions and competing advice from other health providers was reported as further barriers to successful implementation.⁶⁰ Other pragmatic issues included limited internet connectivity,^{32 71} information gathering limitations of telephone and videoconferencing compared with face-to-face appointments,⁴⁹ billing issues^{59 70} and difficulties sustaining participant recruitment and engagement over time.⁷² One study did not achieve its stated aims of changing health outcomes using digital interventions.⁴¹

Some articles presented evidence that interventions were highly feasible. One study described its intervention as 'similar' to existing programmes and having encountered 'no significant problems'^{38 73}; thus, implying a high degree of feasibility. In fact, one article described its intervention as 'surprisingly feasible'.⁴⁶ Other factors suggesting high feasibility included practitioners describing interventions as important³⁵ and worthy of ongoing utilisation,⁴⁸ and that interventions standardised follow-up visits and assisted with information management.⁶⁰

Consumer acceptance

Acceptability to patients

Many studies provided indicators of the acceptability of interventions to patients, with studies indicating high levels of acceptability.^{22 27-30 43 59 60 65 66 70 71 74 75} Several studies provided other indicators of consumer acceptance such as patients' willingness to recommend the intervention to a friend,^{27 61 70} stable engagement was over a 12-month period²¹ and interventions which were rated as good or better compared with previous experiences

or usual care.^{28 30 35 70} Two studies in Australian Indigenous populations; however, moved beyond surveys to inductively explore acceptability. One study identified that home monitoring applications should have culturally inclusive healthcare elements (eg, Indigenous icons, symbols, artworks), should emphasise holistic and personalised healthcare, monitor well-being, and emphasise community connection between patients and healthcare staff, and have options in how the technology is used (eg, mobile or tablet application).⁵³ The other study evaluated telehealth consultations and found they are valued for their usefulness for sharing health concerns, receiving reassurance, obtaining medication prescriptions and for the feeling of connectedness they provided to health service personnel during the COVID-19 pandemic.⁴⁹

DISCUSSION

This systematic review comprehensively assessed 66 studies using an array of interventions, methodologies and different healthcare models. Most studies highlighted benefits to patients derived from rural digital health interventions, spanning a wide range of ICD-10 classifications, with a notable focus on diabetes. The thematic synthesis generated three key themes: cost and time effectiveness, quality healthcare provision and consumer acceptance. These themes suggest that digital health interventions can improve healthcare accessibility and quality for rural residents, are acceptable to rural consumers and have the potential to lower personal and systematic healthcare costs. Taken together, these findings, spanning a wide range of technologies, interventions and countries, suggest that digital health interventions in primary care are highly effective in improving patient outcomes.

The approaches to measuring technology acceptance of interventions lacked rigour and repeatability across studies. Research into technology acceptance in healthcare has shown that the Technology Acceptance Model and unified theory of acceptance and use of technology are well suited to explaining the acceptance of healthcare technologies.^{76 77} Both models estimate behavioural intention which is strongly predictive of actual usage. Future studies exploring the use of technology in rural areas should strongly consider using these, or other similar models, to formally assess and report on patients' acceptance of their intervention. Despite this limitation, policymakers and healthcare sector stakeholders should take comfort that patients from non-homogeneous geography and social contexts generally reported high satisfaction with their interventions: perhaps indicating a general willingness to trial novel approaches that remove barriers to accessing healthcare in rural areas. The most pressing issues potentially limiting digital health interventions are pragmatic policy issues, such as billing, internet connections and technical support, which likely influence increased staff workload, which itself was mentioned as a limitation in several studies. Policymakers must be attentive to the direction of future technological advancements

and design flexible policies to support emerging novel digital healthcare solutions. This includes the seemingly inevitable rise of artificial intelligence in healthcare, which was largely absent from our systematic review results, but one that undoubtedly warrants a targeted systematic review of its own.

Some studies described best practice approaches to healthcare, for instance, team-based interprofessional care. These approaches take a strength-based (rather than deficit-based) approach to the rural health workforce, where local capabilities are supported with digital technology and external expertise to maximise impact on the ground in rural communities. These rural site-specialist assessment models are well accepted in some fields, like optometry, and should be considered more broadly, especially as more generalist roles with lower training time and cost requirements become available within the rural health workforce. In the Australian context, this could include greater utilisation of community health workers, assistants in allied health, and Aboriginal Health Workers and Practitioners to support care delivery across various health domains. Studies from second and third world nations using rural site-specialist assessment models could have direct positive implications for remote communities in first world countries, which often experience third world living conditions. A discussion of the translation of these findings to such communities is beyond the remit of this review, but would be worthy of further primary research.

Costs are an important factor to consider in the rollout of any digital health intervention, noting that 'digital health' is a broad term, and that the complexity of these interventions can vary widely. Similarly, financial outlay and ongoing maintenance costs will vary, and in several studies, were found to be minimal or to even generate savings for patients and healthcare systems. Equally, many interventions did not require extensive information technology skills or assets, making it possible that even low-resourced services could tangibly improve healthcare quality by using digital health interventions. It is important to note, however, that the generalisability of findings about intervention operating costs is low, given that the data presented here are obtained from the USA, Germany and Australia which have substantially different healthcare funding models.

Strengths and limitations

The limitations of this study include the absence of an agreed international definition of 'rurality'. Readers are thus advised to review the included studies to ensure that the described rurality is comparable to their own before seeking to translate findings. An unexpected observation of this study was the negligible appearance of artificial intelligence in the included studies. The literature search strategy does not explicitly exclude such research, suggesting that AI has either made little impact on digital rural health interventions or, more probably, has been used in broader contexts and has not been separately

investigated or reported on in rural and remote populations, which may be considered as a limitation. Future systematic reviews on this topic will be warranted as AI research progresses.

The breadth of this review introduces strengths and limitations to the generalisability of its findings. For instance, its breadth limits the analyses which can be presented to simple descriptive summaries and content analysis. In this regard, though the literature followed the rigorous expectations of a systematic review, the findings are more typical of those presented in a scoping review. Conversely, the same breadth allows the compiling and comparing of a broad range of potentially complementary technologies from different contexts in one place. This increases the generalisability of findings to many health and research fields. Our findings thereby encourage cross-disciplinary learning and increase awareness of the breadth of unique digital health technologies which have been used to improve rural healthcare outcomes. Finally, though many geographical regions are represented in this review, many originate from America (no doubt reflecting the greater capacity for academic output in developed economies). The translation of findings presented in this review may therefore require nuanced consideration, particularly given the substantially different healthcare funding models which exist across the globe.

CONCLUSION

Digital health interventions in rural PHC have been demonstrated to support a broad range of benefits across technology types and social contexts. Most digital health interventions are cost-effective, time-efficient, improve quality healthcare provision and are acceptable to rural and remote healthcare consumers. Digital health technology has been shown to improve healthcare access and health outcomes for rural populations, despite substantial barriers to healthcare access experienced by these communities. Digital health technology has the potential to overcome or reduce some of these barriers, in particular travel cost and time, by providing remote access to higher-level care and regular monitoring. This review provides a global and overarching analysis of primary research studies to clearly highlight that digital health programmes can improve healthcare quality, access and outcomes in rural areas. Digital health technology can underpin viable healthcare models in rural and remote areas and should be prioritised for government funding and investment to assist in closing rural healthcare gaps across the globe.

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REFERENCES

- 1 AIHW. Rural and Remote Health. Canberra: Australian Institute of Health and Welfare, 2024.
- 2 CDC. Leading Causes of Death in Rural America. Rural Health Information Hub, US Centers for Disease Control and Prevention, 2024.
- 3 García MC, Rossen LM, Matthews K, *et al*. Preventable Premature Deaths from the Five Leading Causes of Death in Nonmetropolitan

- and Metropolitan Counties, United States, 2010–2022. *MMWR Surveill Summ* 2024;73:1–11.
- 4 Coombs NC, Campbell DG, Caringi J. A qualitative study of rural healthcare providers' views of social, cultural, and programmatic barriers to healthcare access. *BMC Health Serv Res* 2022;22:438.
 - 5 Buzza C, Ono SS, Turvey C, et al. Distance is relative: unpacking a principal barrier in rural healthcare. *J Gen Intern Med* 2011;26 Suppl 2:648–54.
 - 6 Douthit N, Kiv S, Dwolatzky T, et al. Exposing some important barriers to health care access in the rural USA. *Public Health (Fairfax)* 2015;129:611–20.
 - 7 Maganty A, Byrnes ME, Hamm M, et al. Barriers to rural health care from the provider perspective. *Rural Remote Health* 2023;23:7769.
 - 8 Dassah E, Aldersey H, McColl MA, et al. Factors affecting access to primary health care services for persons with disabilities in rural areas: a “best-fit” framework synthesis. *Glob Health Res Policy* 2018;3:36.
 - 9 Russell DJ, Humphreys JS, Ward B, et al. Helping policy-makers address rural health access problems. *Aust J Rural Health* 2013;21:61–71.
 - 10 Marshall A. A new rural digital divide? Taking stock of geographical digital inclusion in Australia. *Media International Australia* 2024;190:68–84.
 - 11 World health organisation. Global Strategy on Digital Health 2020–2025. Geneva, 2021.
 - 12 Stoumpos AI, Kitsios F, Talias MA. Digital Transformation in Healthcare: Technology Acceptance and Its Applications. *Int J Environ Res Public Health* 2023;20:3407.
 - 13 Shajari S, Kuruvinashetti K, Komeili A, et al. The Emergence of AI-Based Wearable Sensors for Digital Health Technology: A Review. *Sensors (Basel)* 2023;23:9498.
 - 14 Nasir BF, Leedie F, Fox J, et al. Can digital innovation transform rural primary health care? A systematic review protocol. *BMJ Open* 2024;14:e083152.
 - 15 Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;4:1.
 - 16 Hong QN, Pluye P, Fàbregues S, et al. Improving the content validity of the mixed methods appraisal tool: a modified e-Delphi study. *J Clin Epidemiol* 2019;111:49–59.
 - 17 Hong QN, Gonzalez-Reyes A, Pluye P. Improving the usefulness of a tool for appraising the quality of qualitative, quantitative and mixed methods studies, the Mixed Methods Appraisal Tool (MMAT). *J Eval Clin Pract* 2018;24:459–67.
 - 18 Souto RQ, Khanassov V, Hong QN, et al. Systematic mixed studies reviews: updating results on the reliability and efficiency of the Mixed Methods Appraisal Tool. *Int J Nurs Stud* 2015;52:500–1.
 - 19 Hsieh H-F, Shannon SE. Three approaches to qualitative content analysis. *Qual Health Res* 2005;15:1277–88.
 - 20 Finfgeld-Connett D. Use of content analysis to conduct knowledge-building and theory-generating qualitative systematic reviews. *Qual Res* 2014;14:341–52.
 - 21 Kirkland EB, Dericke D, Bays CC, et al. Dissemination of Remote Patient Monitoring: An Academic-Community Primary Care Partnership in South Carolina. *J Public Health Manag Pract* 2023;29:516–24.
 - 22 Siminerio L, Krall J, Johnson P, et al. Examining a Diabetes Self-Management Education and Support Telemedicine Model With High-Risk Patients in a Rural Community. *J Diabetes Sci Technol* 2023;17:1190–7.
 - 23 Kobe EA, Lewinski AA, Jeffreys AS, et al. Implementation of an Intensive Telehealth Intervention for Rural Patients with Clinic-Refractory Diabetes. *J Gen Intern Med* 2022;37:3080–8.
 - 24 Camargo MS, Passos LCS, Mistro S, et al. Improving Access to the Glycated Hemoglobin Test in Rural Communities With Point-of-Care Devices: An Application Study. *Front Med* 2021;8:20211122.
 - 25 Kwak MY, Hwang EJ, Lee TH. Effects of the Physician-Primary-Healthcare Nurse Telemedicine Model (P-NTM) on Medication Adherence and Health-Related Quality of Life (HRQoL) of Patients with Chronic Disease at Remote Rural Areas. *Int J Environ Res Public Health* 2021;18:2502.
 - 26 Lu AD, Gunzburger E, Glorioso TJ, et al. Impact of Longitudinal Virtual Primary Care on Diabetes Quality of Care. *J Gen Intern Med* 2021;36:2585–92.
 - 27 McLendon SF, Wood FG, Stanley N. Enhancing diabetes care through care coordination, telemedicine, and education: Evaluation of a rural pilot program. *Public Health Nurs* 2019;36:310–20.
 - 28 Smith NM, Satyshur RD. Pediatric Diabetes Telemedicine Program Improves Access to Care for Rural Families: Role of APRNs. *Pediatr Nurs* 2016;42:294–9.
 - 29 Shane-McWhorter L, Lenert L, Petersen M, et al. The Utah Remote Monitoring Project: improving health care one patient at a time. *Diabetes Technol Ther* 2014;16:653–60.
 - 30 Siminerio L, Ruppert K, Huber K, et al. Telemedicine for Reach, Education, Access, and Treatment (TREAT): linking telemedicine with diabetes self-management education to improve care in rural communities. *Diabetes Educ* 2014;40:797–805.
 - 31 Egede LE, Dawson AZ, Walker RJ, et al. Randomized controlled trial of technology-assisted case management in low-income adults with type 2 diabetes: Effect on quality of life and blood pressure. *J Telemed Telecare* 2024;30:107–15.
 - 32 Leonard C, Kenney RR, Syed A, et al. Patient perceptions of rural telehealth diabetes control program in Department of Veteran's Affairs. *J Rural Health* 2025;41:e70027.
 - 33 Mackwood M, Pashchenko O, Leggett C, et al. Telehealth Trends and Hypertension Management Among Rural and Medicaid Patients After COVID-19. *Telemedicine and E-Health* 2024;30:e1677–88.
 - 34 Zhang P, Tao X, Ma Y, et al. Improving the management of type 2 diabetes in China using a multifaceted digital health intervention in primary health care: the SMARTDiabetes cluster randomised controlled trial. *The Lancet Regional Health - Western Pacific* 2024;49:101130.
 - 35 Valpuesta Martín Y, Pacheco Callirgos GE, Maroto Martín TM, et al. Satisfaction of patients and primary care professionals with a teleophthalmology-based screening programme for diabetic retinopathy in a rural area in Castilla y León, Spain. *Rural Remote Health* 2020;20:5180.
 - 36 Chauhan V, Negi PC, Raina S, et al. Smartphone-based tele-electrocardiography support for primary care physicians reduces the pain-to-treatment time in acute coronary syndrome. *J Telemed Telecare* 2018;24:540–6.
 - 37 Patel A, Praveen D, Maharani A, et al. Association of Multifaceted Mobile Technology-Enabled Primary Care Intervention With Cardiovascular Disease Risk Management in Rural Indonesia. *JAMA Cardiol* 2019;4:978–86.
 - 38 Doogue R, Hayes P, Hebert R, et al. Towards an integrated blood pressure self-monitoring solution for stroke/TIA in Ireland: a mixed methods feasibility study for the TASMIN5S IRL randomised controlled trial. *Pilot Feasibility Stud* 2023;9:9:9.
 - 39 Zhang X, Wang S, Zhou X, et al. A village doctor-led mobile health intervention for cardiovascular risk reduction in rural China: cluster randomised controlled trial. *BMJ* 2025;389:e082765.
 - 40 Tian M, Ajay VS, Dunzhu D, et al. A Cluster-Randomized, Controlled Trial of a Simplified Multifaceted Management Program for Individuals at High Cardiovascular Risk (SimCard Trial) in Rural Tibet, China, and Haryana, India. *Circulation* 2015;132:815–24.
 - 41 Peiris D, Praveen D, Mogulluru K, et al. SMARTHealth India: A stepped-wedge, cluster randomised controlled trial of a community health worker managed mobile health intervention for people assessed at high cardiovascular disease risk in rural India. *PLoS One* 2019;14:e0213708.
 - 42 Schumacher TL, Jansson A, Herbert J, et al. The effectiveness of medical nutrition therapy for people at moderate to high risk of cardiovascular disease in an Australian rural primary care setting: 12-month results from a pragmatic cluster randomised controlled trial. *BMC Health Serv Res* 2025;25:956.
 - 43 Herbert J, Schumacher T, Brown LJ, et al. Healthy rural hearts: The feasibility of a telehealth nutrition randomised controlled trial for rural people at risk of cardiovascular disease. *J Telemed Telecare* 2025;31:1114–28.
 - 44 Schumacher TL, Clarke ED, Herbert J, et al. The effect of telehealth-based medical nutrition therapy on cardiovascular disease risk factors in a rural population: a secondary analysis of outcomes related to nutrition, health and well-being from the healthy rural hearts randomised controlled trial. *Int J Behav Nutr Phys Act* 2025;22:126.
 - 45 Maharani ASujarwotoPraveen D, et al. Implementation of mobile-health technology is associated with five-year survival among individuals in rural areas of Indonesia. *PLOS Digit Health* 2024;3:e0000476.
 - 46 Norberg BL, Austad B, Kristiansen E, et al. The Dynamics of Doctor-Patient Communication During Remote Consultations: Qualitative Study Among Norwegian Contract General Practitioners. *J Med Internet Res* 2025;27:e57679.
 - 47 Lamas C de A, Santana Alves PG, Nader de Araújo L, et al. Telehealth Initiative to Enhance Primary Care Access in Brazil (UBS+Digital Project): Multicenter Prospective Study. *J Med Internet Res* 2025;27:e68434.
 - 48 Verma N, Buch B, Taralekar R, et al. Diagnostic Concordance of Telemedicine as Compared With Face-to-Face Care in Primary

- Health Care Clinics in Rural India: Randomized Crossover Trial. *JMIR Form Res* 2023;7:e42775.
- 49 Beks H, Mitchell F, Charles J, *et al.* Implementation of telehealth primary health care services in a rural Aboriginal Community-Controlled Health Organisation during the COVID-19 pandemic: a mixed-methods study. *Rural Remote Health* 2023;23:7521.
 - 50 Charly N, Swedlund M. Can You Hear Me Now? Patient Perceptions of Telehealth in a Rural Primary Care Population. *Telemed J E Health* 2024;30:e1719–26.
 - 51 Haggerty T, Stephens HM, Peckens SA, *et al.* Telemedicine versus in-Person Primary Care: Impact on Visit Completion Rate in a Rural Appalachian Population. *J Am Board Fam Med* 2022;35:475–84.
 - 52 Calderon Y, Sandigan G, Tan-Lim CSC, *et al.* Feasibility, acceptability and impact of a clinical decision support tool among primary care providers in an urban, rural and remote site in the Philippines. *BMJ Open Qual* 2024;13:e002526:13.
 - 53 Nasir BF, MacAskill W, Leedie F, *et al.* Co-creating an Indigenous-led virtual health services model for Indigenous Australians living with chronic disease. *Australian J Rural Health* 2025;33:e13206.
 - 54 Mallow JA, Theeke LA, Theeke E, *et al.* The effectiveness of ml SMART: A nurse practitioner led technology intervention for multiple chronic conditions in primary care. *Int J Nurs Sci* 2018;5:131–7.
 - 55 de Battle J, Massip M, Vargiu E, *et al.* Implementing Mobile Health-Enabled Integrated Care for Complex Chronic Patients: Intervention Effectiveness and Cost-Effectiveness Study. *JMIR Mhealth Uhealth* 2021;9:e22135.
 - 56 Wańczura P, Aebischer D, Stecko W, *et al.* Supervised telemedical monitoring in the heart failure population in the regions of social exclusion is associated with improved functional capacity and quality of life. *Pol Arch Intern Med* 2025;135:16954.
 - 57 Reid SC, Kauer SD, Hearps SJ, *et al.* A mobile phone application for the assessment and management of youth mental health problems in primary care: a randomised controlled trial. *BMC Fam Pract* 2011;12:12.
 - 58 Rodríguez Sánchez-Laulhé P, Biscarri-Carbonero Á, Suero-Pineda A, *et al.* The effects of a mobile app-delivered intervention in people with symptomatic hand osteoarthritis: a pragmatic randomized controlled trial. *Eur J Phys Rehabil Med* 2023;59:54–64.
 - 59 Feldmeier G, Löffler C, Altiner A, *et al.* Audiovisual teleconsultation for patients with epilepsy in primary care in rural Germany: a pilot study on feasibility and acceptance. *Pilot Feasibility Stud* 2022;8:213.
 - 60 Gong E, Sun L, Long Q, *et al.* The Implementation of a Primary Care-Based Integrated Mobile Health Intervention for Stroke Management in Rural China: Mixed-Methods Process Evaluation. *Front Public Health* 2021;9:774907.
 - 61 Richter KP, Shireman TI, Ellerbeck EF, *et al.* Comparative and cost effectiveness of telemedicine versus telephone counseling for smoking cessation. *J Med Internet Res* 2015;17:e113.
 - 62 Orchard J, Li J, Freedman B, *et al.* Atrial Fibrillation Screen, Management, and Guideline-Recommended Therapy in the Rural Primary Care Setting: A Cross-Sectional Study and Cost-Effectiveness Analysis of eHealth Tools to Support All Stages of Screening. *J Am Heart Assoc* 2020;9:e017080.
 - 63 Davidson T, Lindelöf A, Wallén T, *et al.* Point-of-care monitoring of warfarin treatment in community dwelling elderly--A randomised controlled study. *J Telemed Telecare* 2015;21:298–301.
 - 64 Yan LL, Gong E, Gu W, *et al.* Effectiveness of a primary care-based integrated mobile health intervention for stroke management in rural China (SINEMA): A cluster-randomized controlled trial. *PLoS Med* 2021;18:e1003582.
 - 65 Meng X, Dai Z, Wang Y, *et al.* Application of Smartphone Otoscope in Telemedicine in Rural Medical Consortium in Eastern China in the COVID-19 Era. *Front Public Health* 2022;10:879410.
 - 66 Shah P, Madhiwala N, Shah S, *et al.* High uptake of an innovative mobile phone application among community health workers in rural India: An implementation study. *Natl Med J India* 2019;32:262.
 - 67 Scogin F, Lichstein K, DiNapoli EA, *et al.* Effects of Integrated Telehealth-Delivered Cognitive-Behavioral Therapy for Depression and Insomnia in Rural Older Adults. *J Psychother Integr* 2018;28:292–309.
 - 68 Marshall SA, Siebenmorgen LE, Youngen K, *et al.* Primary Care Providers' Experiences Treating Opioid Use Disorder Using Telehealth in the Height of the COVID-19 Pandemic. *J Prim Care Community Health* 2024;15:21501319241246359.
 - 69 Zullig LL, Lewinski AA, Woolson SL, *et al.* Research-practice partnerships: Adapting a care coordination intervention for rural Veterans over 3 years at multiple sites. *J Rural Health* 2023;39:575–81.
 - 70 Brown JD, Hales S, Evans TE, *et al.* Description, utilisation and results from a telehealth primary care weight management intervention for adults with obesity in South Carolina. *J Telemed Telecare* 2020;26:28–35.
 - 71 Bonnell S, Griggs A, Avila G, *et al.* Community Health Workers and Use of mHealth: Improving Identification of Pregnancy Complications and Access to Care in the Dominican Republic. *Health Promot Pract* 2018;19:331–40.
 - 72 Brown LR, Sullivan F, Treweek S, *et al.* Increasing uptake to a lung cancer screening programme: building with communities through co-design. *BMC Public Health* 2022;22:815.
 - 73 Davis AM, Sampilo M, Gallagher KS, *et al.* Treating Rural Pediatric Obesity Through Telemedicine: Outcomes From a Small Randomized Controlled Trial. *J Pediatr Psychol* 2013;38:932–43.
 - 74 Barrera TL, Cummings JP, Armento M, *et al.* Telephone-Delivered Cognitive-Behavioral Therapy for Older, Rural Veterans with Depression and Anxiety in Home-Based Primary Care. *Clin Gerontol* 2017;40:114–23.
 - 75 Fortney JC, Pyne JM, Mouden SB, *et al.* Practice-Based Versus Telemedicine-Based Collaborative Care for Depression in Rural Federally Qualified Health Centers: A Pragmatic Randomized Comparative Effectiveness Trial. *AJP* 2013;170:414–25.
 - 76 Adnan A, Irvine RE, Williams A, *et al.* Improving Acceptability of mHealth Apps-The Use of the Technology Acceptance Model to Assess the Acceptability of mHealth Apps: Systematic Review. *J Med Internet Res* 2025;27:e66432.
 - 77 AlQudah AA, Al-Emran M, Shaalan K. Technology Acceptance in Healthcare: A Systematic Review. *Appl Sci (Basel)* 2021;11:10537.
 - 78 Martin YV, Pacheco GE, Martin TMM, *et al.* Satisfaction of patients and primary care professionals with a teleophthamology-based screening programme for diabetic retinopathy in a rural area in Castilla y Leon, Spain. *Rural Remote Health* 2020;20:81–7.
 - 79 Vedanthan R, Kamano JH, DeLong AK, *et al.* Community Health Workers Improve Linkage to Hypertension Care in Western Kenya. *J Am Coll Cardiol* 2019;74:1897–906.
 - 80 Chang W, Homer M, Rossi MI. Use of Clinical Video Telehealth as a Tool for Optimizing Medications for Rural Older Veterans with Dementia. *Geriatrics (Basel)* 2018;3:44.
 - 81 Gurbeta L, Badnjevic A, Maksimovic M, *et al.* A telehealth system for automated diagnosis of asthma and chronic obstructive pulmonary disease. *J Am Med Inform Assoc* 2018;25:1213–7.
 - 82 Shetty R, Samant J, Nayak K, *et al.* Feasibility of Telecardiology Solution to Connect Rural Health Clinics to a Teaching Hospital. *Indian J Community Med* 2017;42:170–3.
 - 83 Evangelista A, Galuppo V, Méndez J, *et al.* Hand-held cardiac ultrasound screening performed by family doctors with remote expert support interpretation. *Heart* 2016;102:376–82.
 - 84 Gonzalez GE, Brossart DF. Telehealth videoconferencing psychotherapy in rural primary care. *Journal of Rural Mental Health* 2015;39:137–52.
 - 85 Maulik PK, Daniel M, Devarapalli S, *et al.* Mental Health Care Support in Rural India: A Cluster Randomized Clinical Trial. *JAMA Psychiatry* 2024;81:1061–70.
 - 86 Alcoceba-Herrero I, Coco-Martín MB, Jiménez-Pérez JM, *et al.* Randomized Controlled Trial to Assess the Feasibility of a Novel Clinical Decision Support System Based on the Automatic Generation of Alerts through Remote Patient Monitoring. *J Clin Med* 2024;13:5974.
 - 87 Chen X, Gong E, Tan J, *et al.* Long-term mortality outcome of a primary care-based mobile health intervention for stroke management: Six-year follow-up of a cluster-randomized controlled trial. *PLoS Med* 2025;22:e1004564.
 - 88 Eberly LA, Tension A, Mays D, *et al.* Telephone-Based Guideline-Directed Medical Therapy Optimization in Navajo Nation: The Hózhó Randomized Clinical Trial. *JAMA Intern Med* 2024;184:681–90.
 - 89 Hickey MD, Owaraganise A, Ogachi S, *et al.* Community health worker-facilitated telehealth for moderate-severe hypertension care in Kenya and Uganda: A randomized controlled trial. *PLoS Med* 2025;22:e1004632.
 - 90 Siedner MJ, Magula N, Mazibuko L, *et al.* Home-Based Care for Hypertension in Rural South Africa. *N Engl J Med* 2025;393:1304–14.